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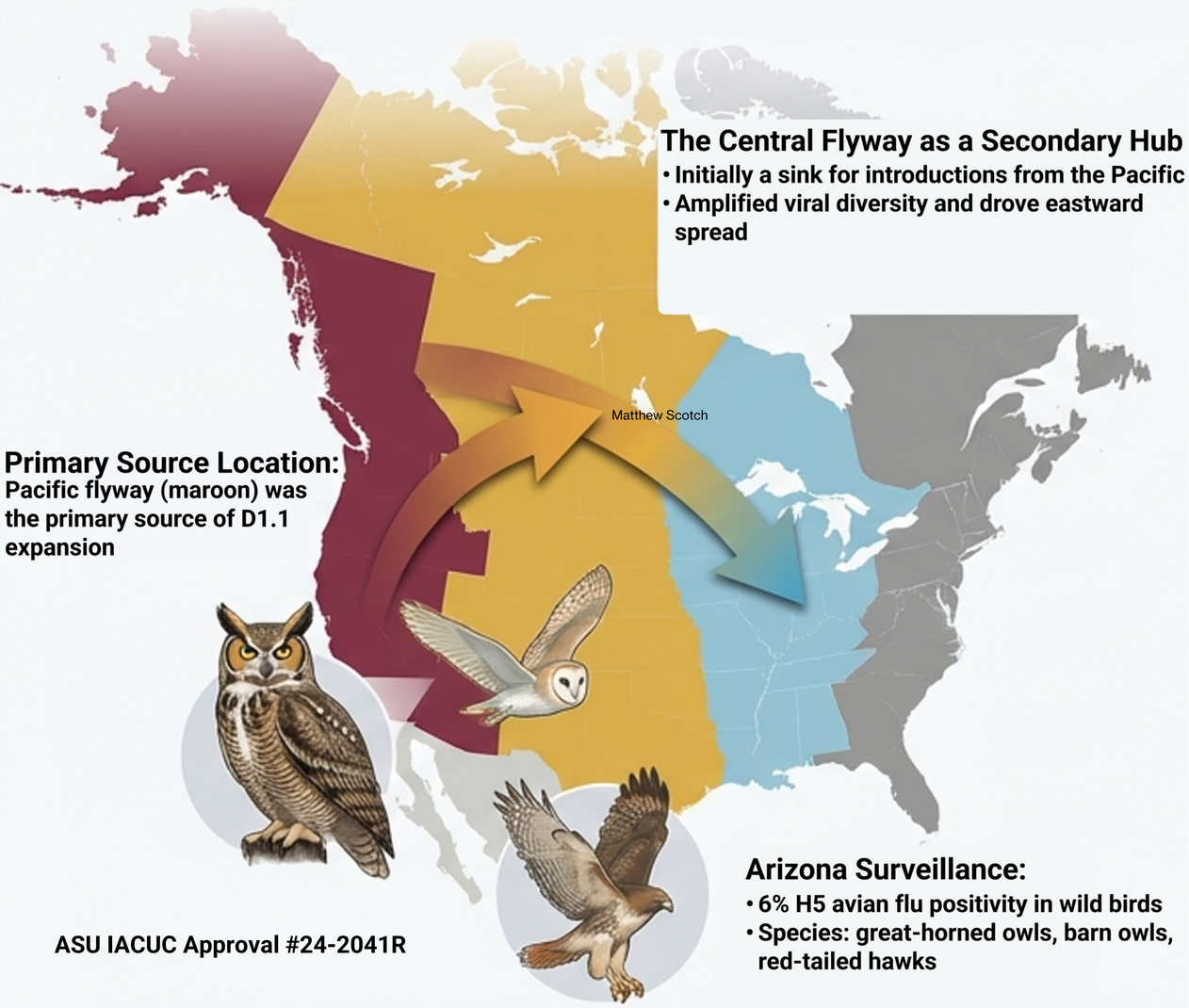
1. College of Health Solutions, ASU; 2. Biodesign Institute, ASU; 3. ASU Health Services; 4. Arizona Game & Fish Department

ASU College of Health Solutions
Arizona State University

Genomic Epidemiology of Influenza A: Tracking Viral Speard from Avian Flyways to Human Communities

Genomic Surveillance Reveals the Hidden Pathways of Influenza Spread in Wild Birds and Humans



The Central Flyway as a Secondary Hub

- Initially a sink for introductions from the Pacific
- Amplified viral diversity and drove eastward spread

Primary Source Location:

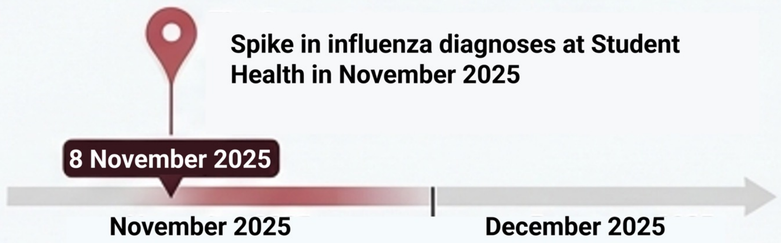
Pacific flyway (maroon) was the primary source of D1.1 expansion

ASU IACUC Approval #24-2041R

Arizona Surveillance:

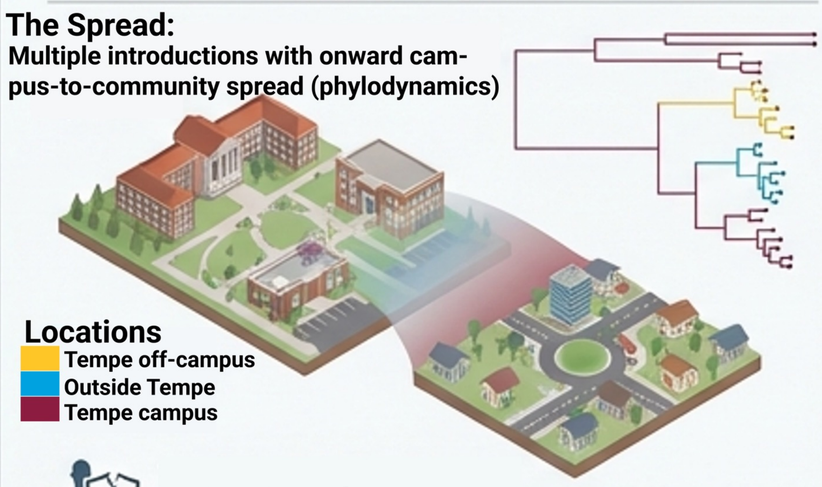
- 6% H5 avian flu positivity in wild birds
- Species: great-horned owls, barn owls, red-tailed hawks

Rapid expansion of Subclade K on campus



The Spread:

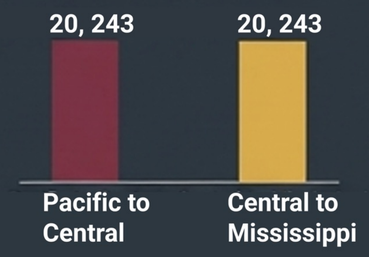
Multiple introductions with onward campus-to-community spread (phylodynamics)



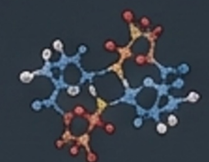
Antigenic Drift:

- 10 subclade K hemagglutinin (HA) substitutions in the RBD and antigenic sites
- Epitope modeling predicts low VE vs. vaccine strain (0.13)

Findings



Decisive Bayes factor support for pairwise transmission



Rapid Evolutionary Rate:
The median rate of evolution for H5 HA was estimated at 7.93×10^{-3} substitutions/site/year

Cryptic Transmission Period:
H5N1 D1.1 likely circulated undetected for 6–16 months

Phylogenetic Clustering by Student Residence:
BaTS shows subclade K is structured by student residence ($p < 0.01$), highlighting social geography in viral mixing.

NotebookLM

Learn about the Scotch lab!

References

1. Scotch et al. Subclade K manuscript. *Euro Surveillance*. 2026
2. Scotch et al. Rapid expansion of HPAI H5 clade 2.3.4.4b genotype D1.1 across North American flyways during fall 2024 avian migration. Under Review.

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